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A STUDY OF CROP YIELD PREDICTION USING REGRESSION TECHNIQUES

by

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Abstract

Agriculture is a core industry that is essential for the sustaining and growth of humans. The increasing populations paired with losses occurring due to various disasters reinforce a need for practices that ensure maximum crop yield. An in-depth analysis of the multiple influencing factors in crop yield will enable farmers to make informed decisions to maximize yield. This project takes a look at three major features namely pesticide usage, rainfall and temperature. It was noted that while pesticide usage initially increased the yield, there is a maximum yield that can be achieved. Hence, the amount of pesticides used should be determined according to each crop and area. A dataset obtained by integrating multiple datasets taken from the FAO and World Data Bank for a time period of 1990 to 2013 is utilized. Once the dataset is explored and the correlations analysed, various regression techniques are implemented to predict crop yield, and compared. The accuracy metrics used were MAE, MAPE, MSE, RMSE, and $R^{2.}$ Among the various ML methods implemented, the Bagging Regressor demonstrated the highest accuracy of 98.59% in yield prediction. A detailed analysis of the influencing factors and their integration into ML techniques could enhance the efficiency and accuracy of the models.

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1 INTRODUCTION

1.1 Background and Context

Food security stands as a critical concern impacting the welfare of populations globally. The Food and Agriculture Organization (FAO) of the United Nations (UN) notes that the need for practices that ensure food security has recently increased as the world is still recovering from a global pandemic, which has further affected the food and energy markets. FAO has stated that there is a need for global food production to grow by 70% to meet increased population needs by the year 2050 (FAO, IFAD, UNICEF, WFP and WHO 2023). Additionally, agriculture serves as the primary economic driver in rural areas, thereby shaping economies. Agriculture is the primary contributor to gross domestic product (GDP), accounting for 6.4% of the global GDP (Shaikh 2022).

Yield prediction, one of the most challenging tasks in precision agriculture, is crucial for governmental entities, agro-industrial enterprises, farmers, and vendors, shaping trading strategies and agricultural planning. It quantifies the yield per unit area over a specified timeframe, integrating diverse factors such as historical production, climate dynamics, soil characteristics, nutrient availability, and pest and occurrence of diseases. Various models and statistical formulations have been developed to predict and analyse crop productivity, facilitating informed decision-making in agriculture (Ramos 2017). With the increasing demand for food production, there is a pressing need for substantial advancements in agricultural practices. Traditional methods of utilizing manual surveys fall short of achieving optimal crop forecasting. The industry is looking for ways where new technology can be utilized to increase harvest and better deal with challenges such as unpredictable weather, population growth, and increased demand for food. Many technological advancements are being utilized for the current model known as “smart agriculture” for enhanced crop yield, processing, storage, and transport. Tools such as using Internet of Things devices to collect and transmit various data, Precision Agriculture techniques that use satellite imagery, GPS technology, and drones to monitor and manage crops with high accuracy, and Robotics & Automation technologies used for tasks such as planting, harvesting, weeding, and spraying are used in the production and harvest. Blockchain technologies are used in the agricultural supply chain. Artificial Intelligence (AI) and Machine Learning (ML) techniques also play a significant role in analysing large and complex datasets to provide insights and predictions. ML techniques are utilized for purposes such as detecting plant diseases and pests, optimizing planting schedules, providing recommendations for custom farming strategies, and predicting crop yield, which is considered for this project.

Multiple ML models have been proposed to predict crop yield, which is one of the more challenging issues in the field. Various factors affect the harvest, including soil conditions, climate,

environment, fertilizer, humidity, etc. Accurate forecasts of crop yields are crucial for managing food security risks. Although process-based crop models offer promising avenues for simulating crop growth at the field level, their application at larger scales introduces sources of error (Hossein Zare 2024).

1.2 Problem definition

Agriculture plays a major role in sustaining and advancing human society. Maximizing yield becomes imperative, particularly amidst the increasing population and various disasters. The study of factors influencing crop yield and the development of predictive models, therefore, represent significant strides towards this objective. This project endeavours to investigate select influencing variables and their impact on yield. Furthermore, it seeks to evaluate various machine learning techniques to discern their efficacy in yielding accurate predictions.

1.3 Aims and Objectives

1.3.1 Forecasting and Analysis of Crop Yield

This project aims to use machine learning models, particularly those using regression techniques to predict crop harvest. Datasets are collected from FAO and World Data Bank and merged. Crop yield is affected by numerous parameters. By implementing several regression models, this project aims to compare to determine which model will give the best results.

1.3.2 Factors Affecting Crop Yield

Crop yield is a multifaceted challenge as the production depends on various factors. This project aims to explore some of the features that affect crop yield and the extent of the inluence. Examining the correlations between influencing factors and the resulting crop yield facilitates the development of efficient predictive models for crop yield estimation.

1.4 Project Structure

The rest of the project is structured in the following manner: Section 2 presents the research conducted on the influencing factors and various ML techniques used in crop yield prediction. Section 3 explains the method followed in preparing, analysing and implementing regression models. Section 4 discusses the evaluation methods used. Section 5 provides the evaluations and analyses of the findings. Section 5 concludes while recognising areas for future works.

1.5 Challenges

With technological advancements, various tools are used to increase the efficiency of crop yield prediction and optimize production. The major challenge faced is the multiple influencing factors and the large uncertainties associated with them. Many of these influencing factors such as weather conditions, rainfall, and temperature are difficult to predict accurately. Therefore, building an efficient model that can consider these challenging factors is not easy.

Modeling complex phenomena with spatiotemporal variations is difficult due to factors like disturbances, modeling errors, and uncertainties in real-world systems. This challenge is compounded when there is missing data in the input dataset (Gavahi 2021).

2 LITERATURE REVIEW

2.1 Background

Machine Learning (ML) and Deep Learning (DL) models have been widely employed for crop yield prediction, leveraging techniques such as regression trees, random forests, artificial neural networks, support vector machines (Kim 2016), convolutional neural networks, decision trees (Kim 2016), and recurrent neural networks. Despite their utility, challenges persist, including lengthy training times, sensitivity to model structure and parameter calibration, and difficulty in explaining model predictions. However, advancements continue to be made, with studies demonstrating the efficacy of algorithms like XGBoost, CNN, and LSTM (N. Kim 2019) (J. You 2017) in improving prediction accuracy across diverse agricultural contexts. While traditional methods have provided valuable insights into crop yield prediction, the integration of Machine Learning and Deep Learning techniques offers a promising avenue for enhancing predictive accuracy and scalability. By harnessing the power of these advanced methodologies and addressing associated challenges, researchers can contribute to more robust and efficient food security management strategies in an ever-changing agricultural landscape.

2.2 Data Science and Agriculture

Climactic changes across the globe in the past years have been rapid and weather conditions are getting increasingly unpredictable. Surface temperatures have been rising, and monsoons are observed to have major fluctuations. Hence, building ML models for agricultural systems is a complex challenge because of its dynamic, non-linear nature. Challenges such as inadequate data collection tools and gaps in knowledge make the process even harder. Regression-based models are implemented for continuous variables such as predicting crop yield and climate patterns. The use of classification models helps achieve objectives such as evaluating the quality of yield and analysing soil properties (Dhivya Elavarasan 2018).

A study was conducted to examine the impact of data analytics on agriculture. (N. Chergui 2020). The authors highlight that crop yield is influenced by multiple factors, and there is further variability resulting from the types of devices used to gather data. Based on these, data is divided into (a)historical data, (b)sensor data, (c)image data or (d)satellite data. Two trends in crop estimation are identified: the first pertains to data sources directly impacting crops (such as soil data, weather data, and environmental parameter data), commonly employed to provide pre-season estimates of crop yield (S. Gyamerah 2020) (L. Kouadio 2018). The second trend involves data sources gathered through advanced technologies and tools like multi-spectral and hyper-spectral satellite images, remote sensing, and sensors. These methods are typically utilized to offer in-season

predictions of crop yields (M. Maimaitijiang 2020). Data mining techniques have multiple applications in agriculture. Some of them are represented in Figure 1 below.

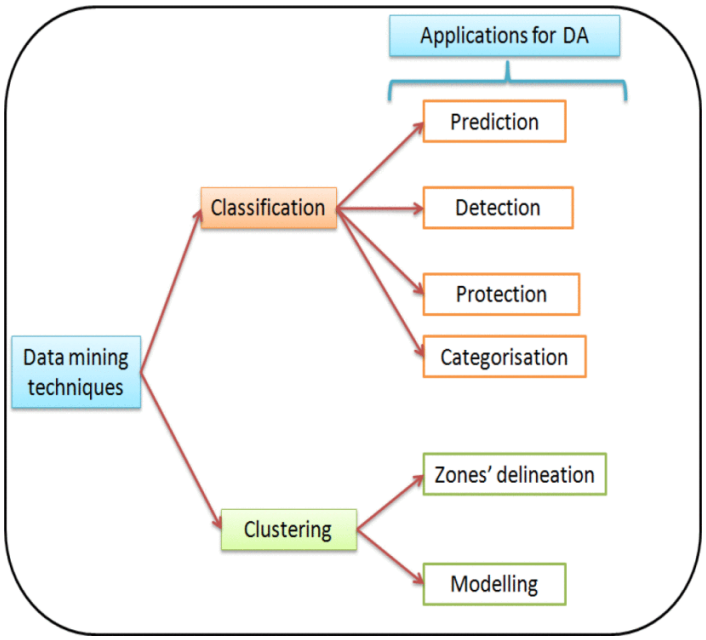


Figure 1: Applications of DM techniques in agriculture (N. Chergui 2020)

2.3 Factors Influencing Crop Production

Crop cultivation and the harvest obtained depends on numerous factors. Climactic and environmental factors such as humidity, temperature, and rainfall influence crop yield. Other factors are soil nutrition/soil content, seed quality, pesticides or diseases. The end goal is to achieve maximum harvest, which can be achieved by studying the correlation of each factor on yield.

2.3.1 Soil Fertility or Composition

Plants depend on good soil for growth. Soil consists of numerous nutrients which help plant growth. This includes both microelements (boron (B), manganese (Mn), zinc (Zn), iron (Fe), molybdenum (Mo), copper (Cu), and chlorine (Cl)), and macronutrients (nitrogen (N), phosphorous (P), sulphur (S), potassium (K), calcium (Ca), and magnesium (Mg)). Deficiency of one or more nutrients or excess of others leading to toxicity leads to an imbalance in soil nutrition and causes a decline in yield and/or quality of produce (Lal 2009).

Soil conditions like drought, mulching, and heat also influence plant growth and crop yield. Fahad (2015) conducted a study on the effects of stress, particularly phytohormones, in plants. Osmotic stress is induced on plants by factors such as drought, freezing or high salinity. These influence growth and yield such as life cycle, shoot growth inhibition or enhanced root growth (Xiong 2002). A study conducted by Feng (2017) concluded that the root compensation effect was directly proportional to the salinity difference in the root zone. The study was done to determine the effects

of using saline or brackish water for irrigation, on the occasion of lack of freshwater. The roots were divided into two, one fed freshwater and the other using brackish water.

Kumar (2022) presented a method for analyzing soil data to classify its suitability for sugarcane cultivation based on parameters such as pH, nitrogen, phosphorus, potash, manganese, copper, and sulfur content. By assessing soil characteristics like pH, electrical conductivity (EC), and organic carbon (OC) levels, the system guides farmers in adjusting soil conditions and nutrient management practices to optimize crop growth. This approach not only aids in identifying nutrient deficiencies but also provides tailored recommendations for sustainable farming practices, contributing to soil conservation and future agricultural sustainability (Kumar 2022).

2.3.2 Pesticide usage

An important aspect of maximising yield is to ensure the crops are protected from harmful external factors. Pesticides - the group consisting of materials used as insecticides, fungicides, rodenticides, herbicides, rodenticides, molluscicides, and nematicides – play a significant part in reducing the losses of produce as well as improving yield quality and quantity. There are multiple ways to classify pesticides. This can be done by functional groups (e.g. insecticide works on insects, herbicides are used to kill weeds), chemical classes (comprised of organic or inorganic ingredients), modes of action, and toxicity (Garcia 2012). While the usage of pesticides can contribute to maximizing agricultural yield, excessive usage poses potential risks to environmental integrity and can result in enduring adverse consequences on soil and water quality (Tudi 2021).

A review conducted by Popp (2013) underscores the role of pesticides in enhancing crop yield. While environmental and health concerns exist, pesticide usage is still increasing. Methods such as Integrated Pest Management (IPM) and utilising natural enemies against pests are helping to reduce the amount of pesticides used. The study notes that determining the correlation of pesticide usage to yield is challenging as other variables also affect the yield.

2.4 Usage of ML Techniques for Crop Yield Forecast

A Systematic literature review on crop yield prediction utilizing ML techniques was done by Thomas Van Klompenburg (2020). Thomas et al. note that the task is not a simple one and involves multiple complex steps as well as numerous datasets. This is due to crop yield depending on many factors such as climate, weather, soil, fertilizer usage, and variety of seed. Since ML techniques can find patterns and correlations, gaining knowledge from these complex and large datasets is a practical approach. It is discussed that while many models can predict reasonably, an approach that performs better and can handle the data volume is desirable. Thomas et al. note that Neural Networks (NN) and Linear Regression Algorithms are the 2 most used algorithms, followed by Random Forest (RF) and Support Vector Machines (SVM). It is noted that Deep NN algorithms with additional and different hidden layers are also used. These may be a convolutional layer or

pooling layers. Another widely used deep learner is noted to be the Long-Short Term Memory (LSTM) Network. These are designed specifically for sequence prediction problems.

Utilizing deep learning algorithms for rapid and precise prediction of crop yield has been studied and compared with the traditional agriculture cycles (Priti P Jorvekar 2023). It is discussed that the accuracy of the forecast is dependent on the relevancy of the features selected. Amongst the deep learners, DNN algorithms with hidden convolution and pooling layers are noted to be among the most used. This is followed by Convolution Neural Network (CNN) and recurrent NN, notably LSTM. Various studies demonstrated that Convolutional Neural Networks (CNNs) excel in analysing spatial features, while Long Short-Term Memory (LSTM) networks are proficient in capturing phenological characteristics. A study done on soybean yield prediction for both in-season and end-of-season combined CNN-LSTM was proposed (Sun 2019). The crop was planted in 31 US states for a time period of 2003 to 2015. MODIS SR data, MODIS LST data, and Daymet weather data were the influencing factors considered and the yield was the label data. A histogram-based approach able to provide more information from the pixel distribution was chosen over the common approach of selecting the mean value or VI over the region. The model was evaluated with RMSE and Percent Error (PE), which showed that the new CNN-LSTM model outperformed the traditional CNN and LSTM.

A research work using data from Manipur, India which included rainfall, temperature, and relative humidity was conducted. The research included 10 different crops for the period of 1997 - 2016 (Bhattacharyya 2021). A regression model was implemented for the dependent variable yield. The R^2 values were calculated for each of the ten crops to explain the dependency of yield on the factors taken as the input. Amongst them, turmeric, chilli and rice showed high R-squared values indicating high dependency. Rainfall was found to have the most influence on the yield in all the crops.

Random Forest has been used for crop yield prediction as well. A study on cotton cultivation was conducted in the state of Maharashtra, India between 2001 and 2017. Data were collected for precipitation, weather conditions, condition of vegetation in response to weather (known as Vegetation condition index or VCI), and heat unit called Growing degree days (GDD). A Random Forest modelling approach to estimate yield for three separate harvestings was performed and was found successful. The performance showed good accuracy, observed by the R^2 values of 0.69 in the final of the three cultivation periods (Prasad 2021).

A new predictive model which integrated ML techniques, agronomic insights and optimization algorithms was presented by Ansarifar (2021). They referred to it as the Interaction Regression model. The study ventured to address the various challenges faced in crop yield prediction techniques. Ansarifar noted that ML models such as regression models, or Neural Networks were

based on environmental and managerial variables alone. The lack of public availability of genotype data was a hindrance. The accuracy of these models is hence susceptible to model structure and parameter calibration. An alternate approach is to use nonlinear models such as APSIM or DSSAT, which use the physiological aspects of plants and soil. Explicit and complex reasoning of the interaction between characteristics and environmental conditions in the lifecycle of crops can be explained in these crop models. Though these models are capable of explaining the complex correlations and hence provide valuable insights, the prediction accuracy is often low. Therefore, Ansarifar proposed a new model which combined the strengths of the two types of yield prediction approaches. In a dataset of three Midwest states in the USA(Illinois, Indiana, and Iowa) from 1990 to 2018, the new model outperformed state-of-the-art ML algorithms, achieving a relative root mean square error of 8% or less. It was also able to identify about 12 interactions and was able to quantify the effect of weather, soil, and management on crop yield.

3 METHODOLOGY

3.1 Data Preparation

The FAO and World Data Bank have collated data on various aspects of Agriculture for monitoring, analysis, and develop models to increase food production. This ensures the major objective of global food security. The dataset used in this project is from data collated from FAO and World Data Bank. Data sets for temperature, rainfall, amount of pesticides used, and yield obtained are integrated and consequently utilized for implementing the various ML techniques.

The yield dataset, collected from FAO, shows the sample data below in Figure 2. The yield dataset was observed to have no Null values.

```
yield_data.sample(5)
```

	Domain Code	Domain	Area Code	Area	Element Code	Element	Item Code	Item	Year Code	Year	Unit	Value
39881	QC	Crops	170	Peru	5419	Yield	125	Cassava	1986	1986	hg/ha	108513
1028	QC	Crops	7	Angola	5419	Yield	116	Potatoes	1992	1992	hg/ha	40000
11756	QC	Crops	44	Colombia	5419	Yield	27	Rice, paddy	1973	1973	hg/ha	40427
33305	QC	Crops	145	Micronesia (Federated States of)	5419	Yield	27	Rice, paddy	2000	2000	hg/ha	11573
6550	QC	Crops	27	Bulgaria	5419	Yield	236	Soybeans	2010	2010	hg/ha	22717

Figure 2: Sample of Yield dataset

This dataset contains the areas where the crops are cultivated, the type of crop, the year of cultivation, and the yield in hectogram per hectare (hg/ha). The value of 1 hectogram is 100 grams.

The next data set obtained from FAO was on **pesticides**, with the following sample set as shown in Figure 3. It was observed that this dataset does not contain any Null values.

	Domain	Area	Element	Item	Year	Unit	Value
3738	Pesticides Use	Syrian Arab Republic	Use	Pesticides (total)	1996	tonnes of active ingredients	540.77
582	Pesticides Use	Brunei Darussalam	Use	Pesticides (total)	2011	tonnes of active ingredients	9.06
2120	Pesticides Use	Kenya	Use	Pesticides (total)	1998	tonnes of active ingredients	4114.00
4228	Pesticides Use	Venezuela (Bolivarian Republic of)	Use	Pesticides (total)	2006	tonnes of active ingredients	3928.00
1787	Pesticides Use	Hungary	Use	Pesticides (total)	2014	tonnes of active ingredients	8970.90

Figure 3: Sample of Pesticides dataset

From the Word Data Bank, data on **temperature** was collected. The sample dataset is shown below in Figure 4. The average column of this dataset was noted to have null values. To fill in the values, more information is needed.

```
temp_data.sample(5)
```

	year	country	avg_temp
54647	1885	Pakistan	21.28
16998	1858	Argentina	16.22
50131	1951	Pakistan	18.40
15348	1984	United States	10.66
3330	1764	Turkey	NaN

Figure 4: Sample of Temperature dataset

The final dataset, which was obtained from the World Data Bank, is **the average rainfall data** (Figure 5). It was observed that the average rainfall column contained Null values in this dataset. Additional information is required to fill in the missing values.

```
rainfall_data.sample(5)
```

	Area	Year	average_rain_fall_mm_per_year
2529	Guinea	2005	1651
3128	Kazakhstan	2015	250
91	Algeria	2016	89
3571	Lithuania	1992	656
3414	Lebanon	1990	661

Figure 5: Sample of rainfall dataset

As is evident from the samples, all four datasets contain the **columns Year and Area**. Hence, these were selected as the **keys for merging the datasets**.

The two common columns were analysed further for efficient data merging. Firstly, the Year column was studied in the 4 datasets. It was observed that the yield dataset had data from the years 1961 to 2016. The temperature dataset was from the year 1743 to 2013. The rainfall dataset contained data from the years 1985 to 2017. And finally, pesticides contained data from 1990 to 2016. It was determined that all the datasets had data present in the years from **1990 to 2013**, giving us 24 years of data.

The other common column was Area. This is the list of countries where the crops are cultivated. The columns were explored for each of the 4 datasets. Initially, only about 50% of yield data had the same area values as the rest of the datasets. It was observed that these were names of

countries which had a name change due to varying reasons. For the current project, it was decided to remove the mismatched countries.

3.1.1 Data Cleaning and Merging

Before merging the four datasets, each was analysed further in detail. It was observed that in the Yield and Pesticides dataset, there were a few columns which had information irrelevant to our use case. These were the columns: Domain, Domain Code, Element, Element Code, and Unit. Additionally, the columns Area Code, Item Code, and Year Code were duplicated information of the columns Area, Item and Year respectively. To eliminate bias in the models, it was decided to remove the aforementioned columns. After cleaning the datasets, they were merged to get the dataset named 'FinalCropYieldData'. This is the final dataset on which the models are tested.

3.1.2 Overview of Dataset

The final dataset contains 7 columns. A sample is shown in Figure 6 below. The Item and Area columns are of the object data type in the dataset. The remaining columns are either integers or floating datatypes. There aren't any missing values in this final data set.

	Area	Item	Year	Yield	Avg_Rain	Pesticide_Amt	Avg_Temp
0	Albania	Maize	1990	36613	1485	121.00	16.37
1	Albania	Potatoes	1990	66667	1485	121.00	16.37
2	Albania	Rice, paddy	1990	23333	1485	121.00	16.37
3	Albania	Sorghum	1990	12500	1485	121.00	16.37
4	Albania	Soybeans	1990	7000	1485	121.00	16.37
...	...	...	...	...	...	...	...
28243	Zimbabwe	Rice, paddy	2013	22581	657	2550.07	19.76
28244	Zimbabwe	Sorghum	2013	3066	657	2550.07	19.76
28245	Zimbabwe	Soybeans	2013	13142	657	2550.07	19.76
28246	Zimbabwe	Sweet potatoes	2013	22222	657	2550.07	19.76
28247	Zimbabwe	Wheat	2013	22888	657	2550.07	19.76

28248 rows × 7 columns

Figure 6: Sample data of the final crop yield dataset

The information contained within is described in Table 1 shown below.

Variable	Description
Area	Name of the countries
Item	Types of crops cultivated
Year	Time of crops cultivated within the years 1985-2013
Yield	Yield obtained for the crop (Measured in hectogram per hectare (Hg/Ha))
Avg_Rain	Average rainfall per year
Pesticides_Amt	Pesticides used in tonnes
Avg_Temp	Average temperature

Table 1: Table describing the columns in the dataset

The features with numerical values that the models use are crop yield (in column Yield), the average rain in millimetres per year (column Avg_Rain), the amount of pesticides used in tonnes (Pesticides_Amt), and the average temperature of the area the crop was cultivated (Avg_Temp). Extracting the features gives us the following information (Figure 7).

```
df.describe().T
```

	count	mean	std	min	25%	50%	75%	max
Year	28248.0	2001.541950	7.052997	1990.00	1995.00	2001.00	2008.00	2013.00
Yield	28248.0	77047.863282	84950.194454	50.00	19918.75	38295.00	104598.25	501412.00
Avg_Rain	28248.0	1148.973272	709.762347	51.00	593.00	1083.00	1668.00	3240.00
Pesticide_Amt	28248.0	37069.136973	59954.787836	0.04	1695.71	17517.76	48687.88	367778.00
Avg_Temp	28248.0	20.543722	6.311828	1.30	16.71	21.51	26.00	30.65

Figure 7: Numerical Features

It is observed that the average rainfall per year overall is approximately 1149, with the least rainfall being 51 and the highest 3240. The average temperature is approximately 21, ranging from 1.30 to 30.65. The pesticide usage showed a mean of 37069 tonnes, with a minimum of only 0.04 to a maximum of 367778 tonnes. Finally, the yield obtained ranged from 50hg/ha to a maximum of 501412 hg/ha, giving an average yield of 77047 hg/ha.

The categorical summary was also explored. In the 28248 sets of data, it was observed that there are 101 unique areas/countries. Among them, India had the highest number of occurrences – 4048 times out of the 28248 entries. The dataset explored 10 different crops: Maize, Potatoes, Rice, Paddy, Sorghum, Wheat, Cassava, Sweet potatoes, Soybeans, Yams, and Plantains and others. Potatoes was the most common crop in the dataset.

3.2 Exploratory Data Analysis

The dataset was further explored to analyse the correlations. A heatmap visualization of the correlation matrix is given below in Figure 8.

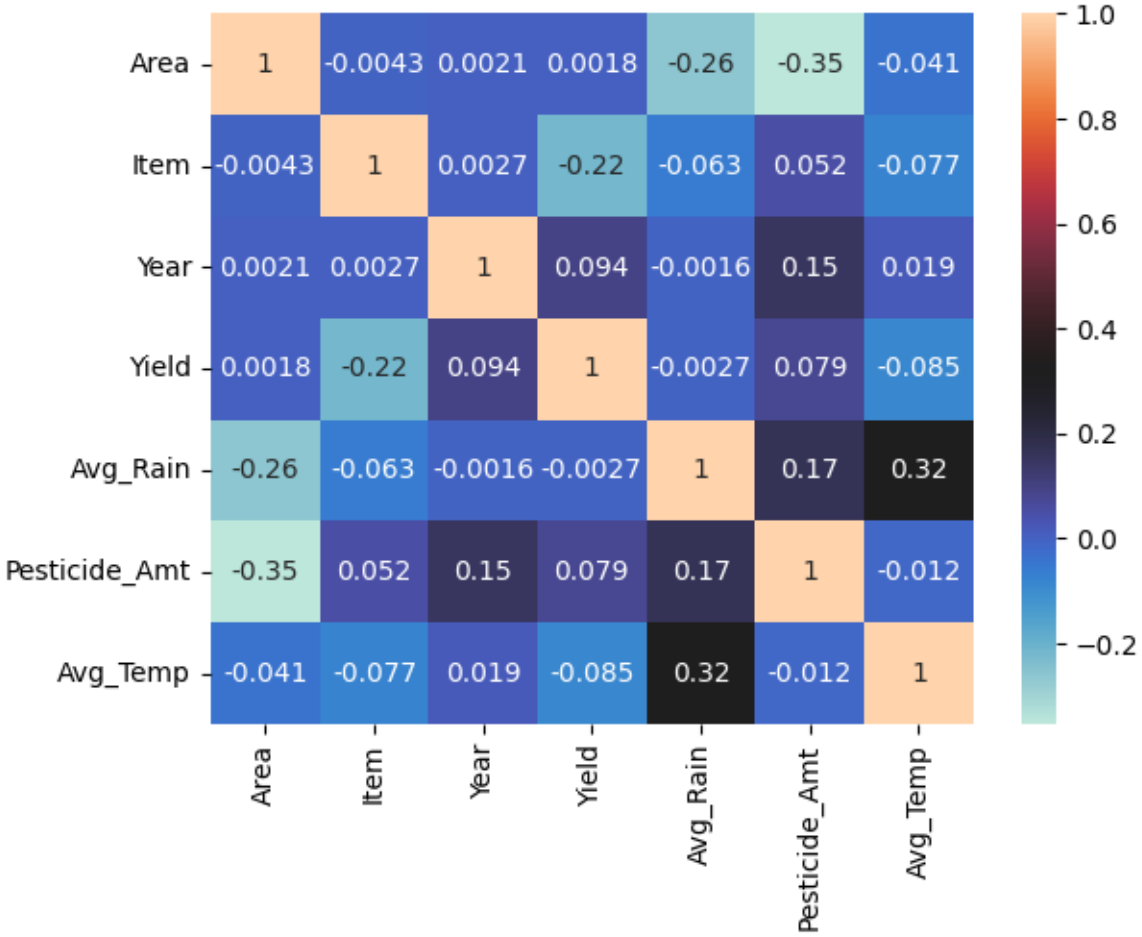


Figure 8: Correlation Heatmap

It was observed that the amount of pesticides used (Pesticide_Amt) and the average rainfall (Avg_Rain) had a strong correlation with the Area. The Item and amount of pesticide used also showed a strong correlation.

Information on variations was explored as well. The bar chart is given below in Figure 9.

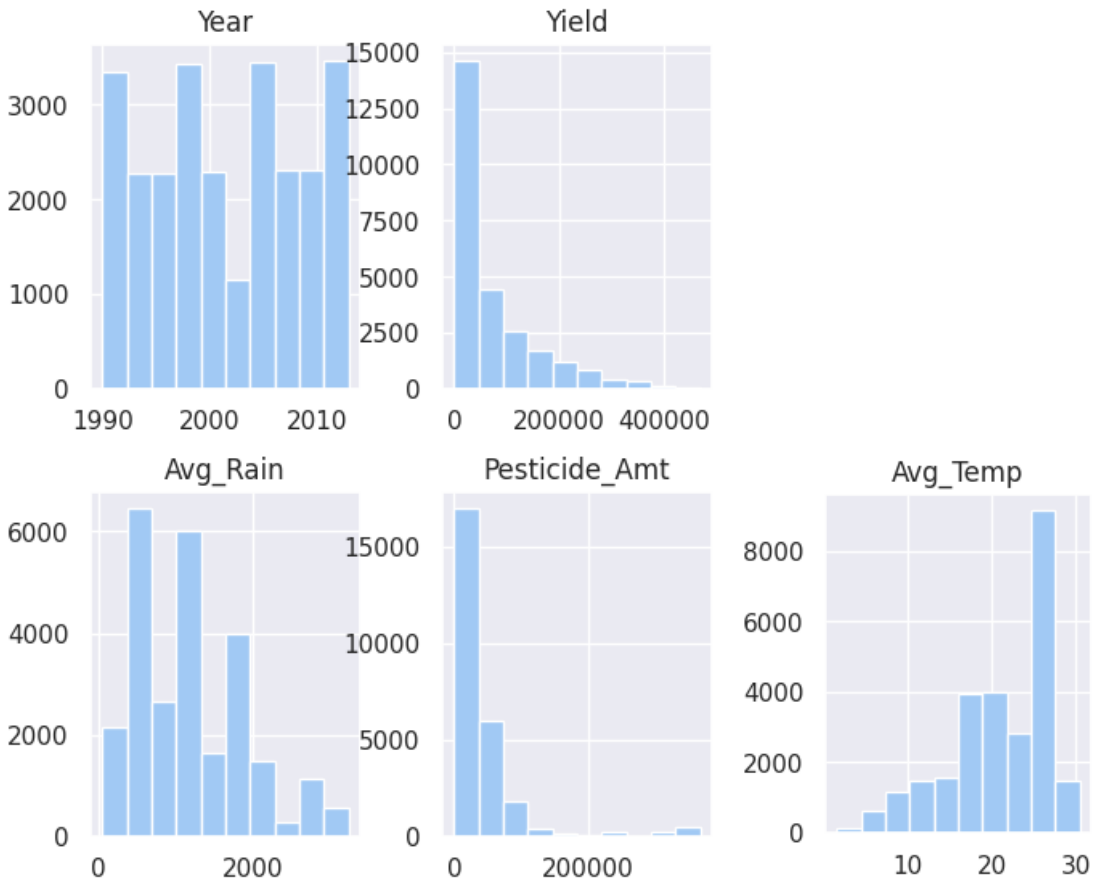


Figure 9: Bar charts for Analysis

It was noted that the Average rain peaked just above 6000. The maximum pesticide usage was around 18000 and the maximum temperature reached 25.

A scatterplot was generated to explore the correlation of each Item/crop cultivated with the various features as shown in Figure 10 below.

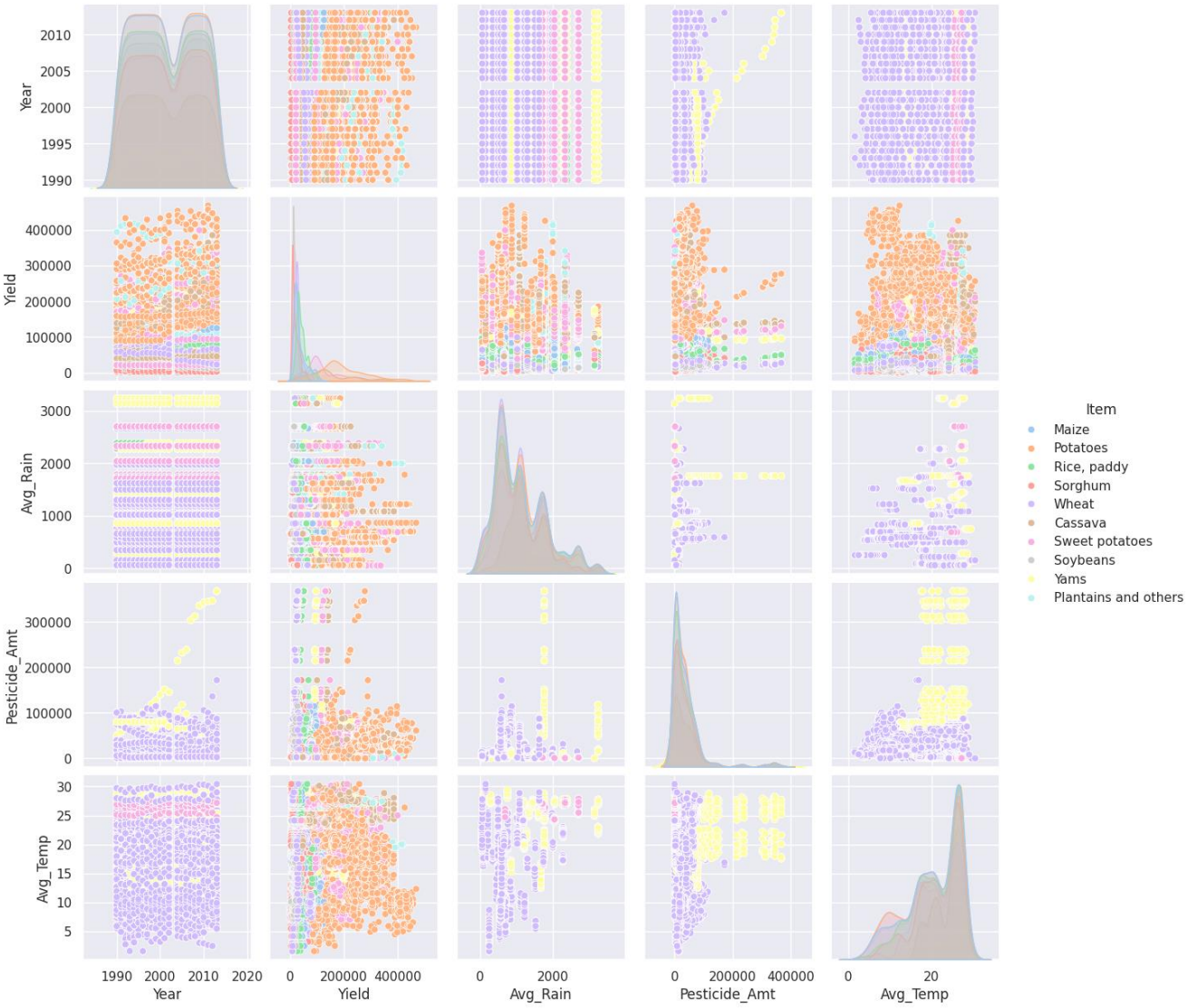


Figure 10: Scatterplot for Correlations with each unique Item

The crop Yams appeared to have higher pesticide usage while growing on all ranges of rainfall. Pesticide usage was noted to increase from 2003 to 2013 as well. Wheat thrived across all temperatures and required average to minimal rainfall. Potatoes had the highest yield amount whereas Sorghum and Soybeans showed to have the least.

In the next step, the Item/crop cultivated was looked into for each Area/country. The countries were divided into chunks for ease of understanding. A set of histograms to represent the distribution of yield values for each subset was plotted and is shown in Figure 11.

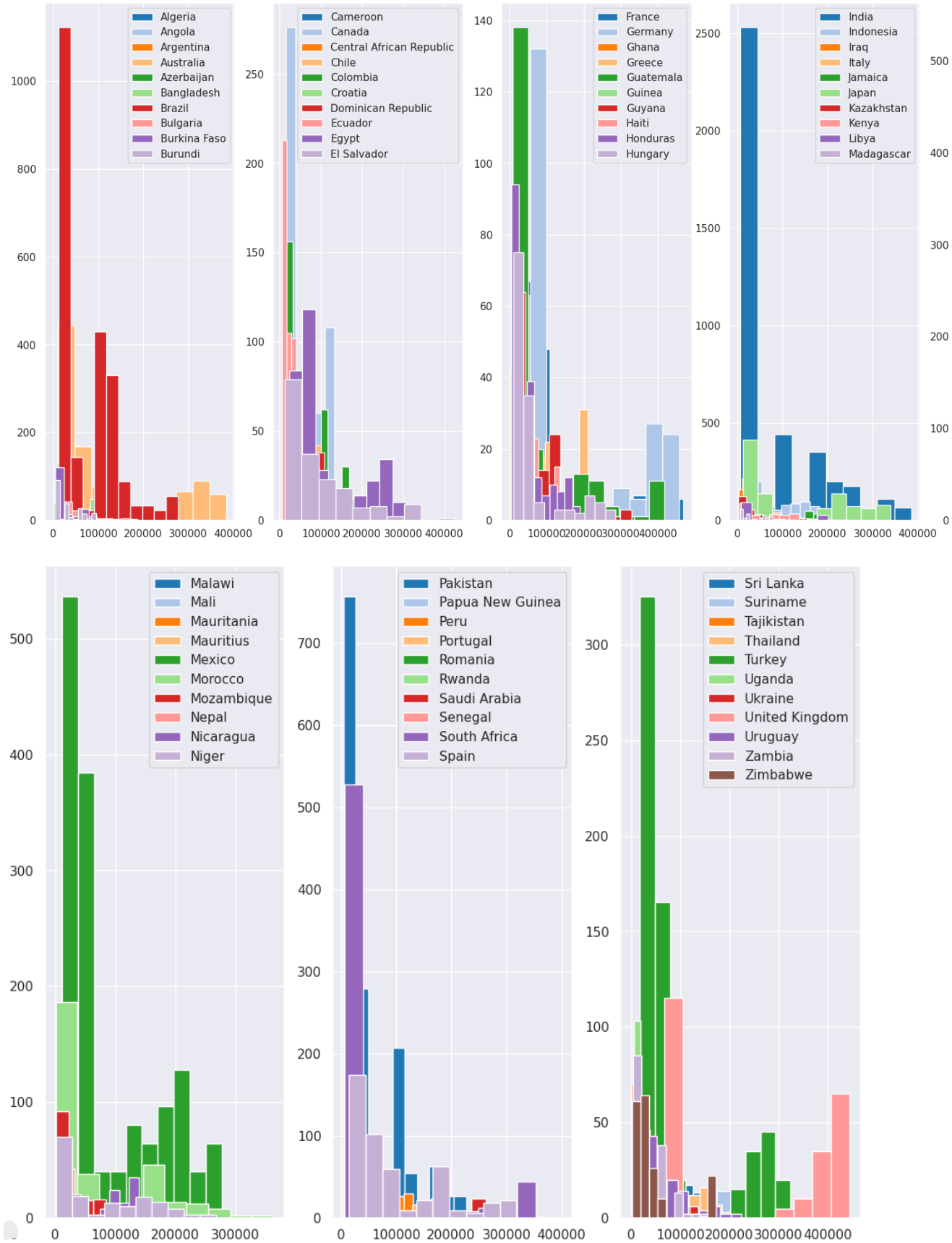


Figure 11: Plot for Yield per Area

The next analysis was done to find the highest yield across the unique Areas and for each Item. Each combination of Area and Item was explored. These were split into groups for ease, and then the lowest and highest yields were calculated. It provided insight into which Area produced the highest yield for which Item. Figure 12 below shows a sample of the findings.

Area	Item	Yield
United Kingdom	Potatoes	406131.695652
France	Potatoes	395037.260870
Germany	Potatoes	386553.347826
Australia	Potatoes	330944.826087
Japan	Potatoes	311005.260870
...	...	...
Zimbabwe	Sorghum	3864.956522
Azerbaijan	Sorghum	3324.166667
Angola	Sorghum	2926.769231
Niger	Sorghum	2592.913043
Tajikistan	Soybeans	941.750000

495 rows × 1 columns

Figure 12: Sample table showing the highest yield per Area and the Item

The top-producing Area/countries were the United Kingdom, France, Germany, Australia and Japan. The crop produced (Item) was potatoes for all these countries. Soybeans and Sorghum were the Items with the least yield and the Areas of cultivation were Angola, Niger and Tajikistan.

Another major factor that determined the crop yield was the average rainfall received. In the next step, the average rainfall per mm (Avg_Rain) for each Area was plotted. This is shown in Figure 13 below.

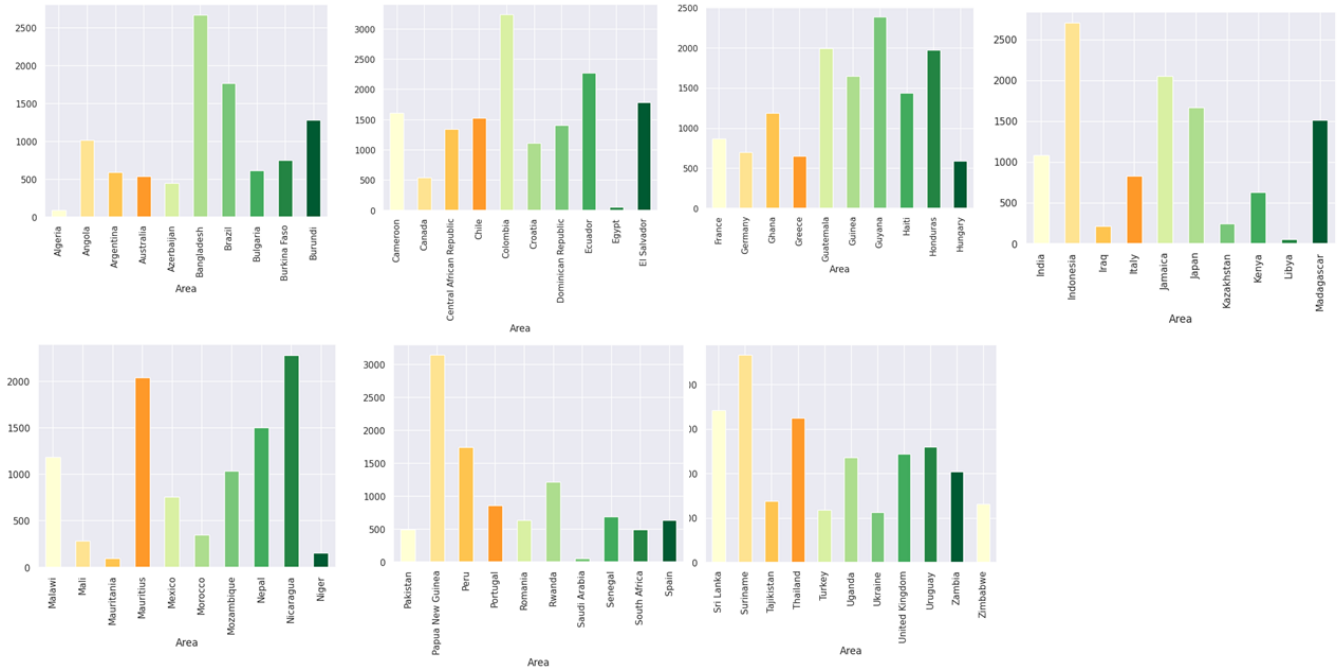


Figure 13: Average Rainfall per Area

The countries with the highest average rainfall were Bangladesh, Columbia, Guyana, Indonesia, Nicaragua, Papa New Guinea, and Suriname with over 2500 mm per area of rainfall. The countries with the least average rainfall of less than 500mm were Azerbaijan, Egypt, Iraq, Kazakhstan, Libya, Mali, Mauritania, Morocco, Niger, Saudi Arabia and South Africa.

Another major factor on which crop yield depends is the use of pesticides. The usage per area was explored next. This is depicted in Figure 14 below. The countries with the most amount of pesticides used were Brazil, Argentina, Italy and France. The areas Angola, Burkina Faso, Burundi, Central African Republic, Ghana, Guinea, Guyana, Haiti, Madagascar, Malawi, Mali, Mauritania, Nepal, Niger, Papua New Guinea, and Uganda were among the least pesticide used. It was noted that all the areas with the lowest pesticide usage showed poor yield. The top producing countries, Germany, Australia, and the United Kingdom all had an average amount of 30000 tonnes of pesticide usage.

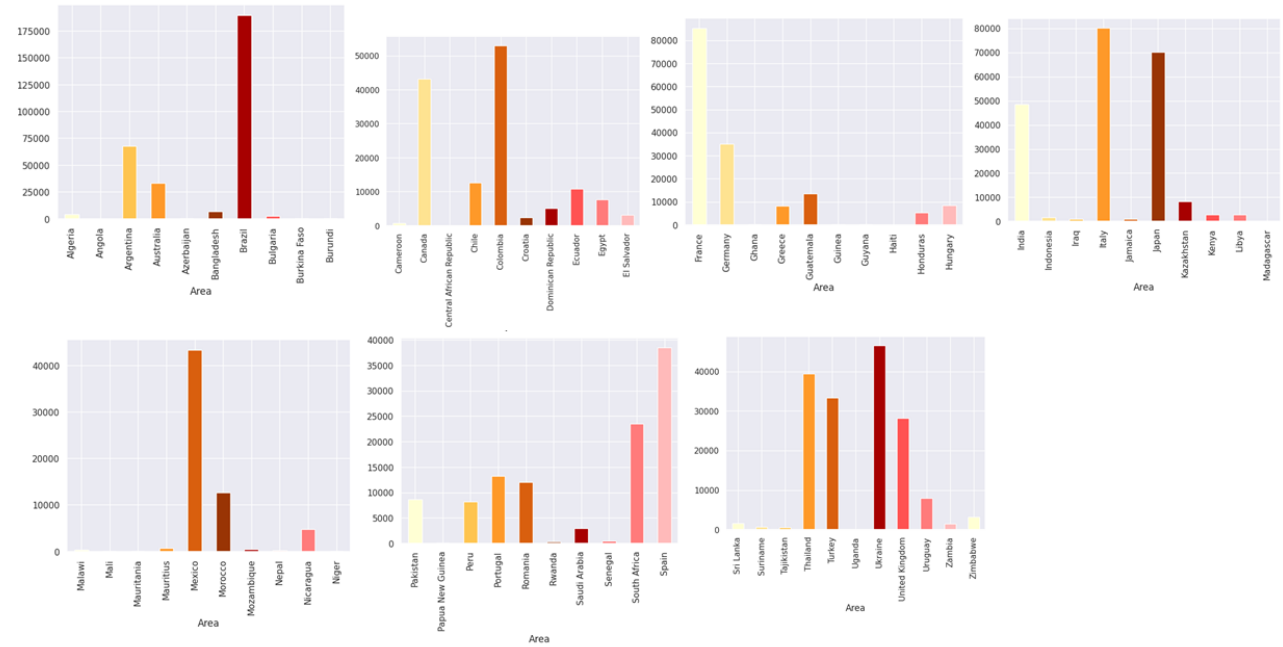


Figure 14: Pesticide amount used per Area

To determine the degree of effect pesticide usage on the crop yield, this correlation was further explored. A set of bar plots showing average values for pesticide usage and yield across each unique area was charted. This is shown in Figure 15 below. A scatterplot that represents the relationship between Yield and Pesticide_Amt was plotted as well, shown below in Figure 16.

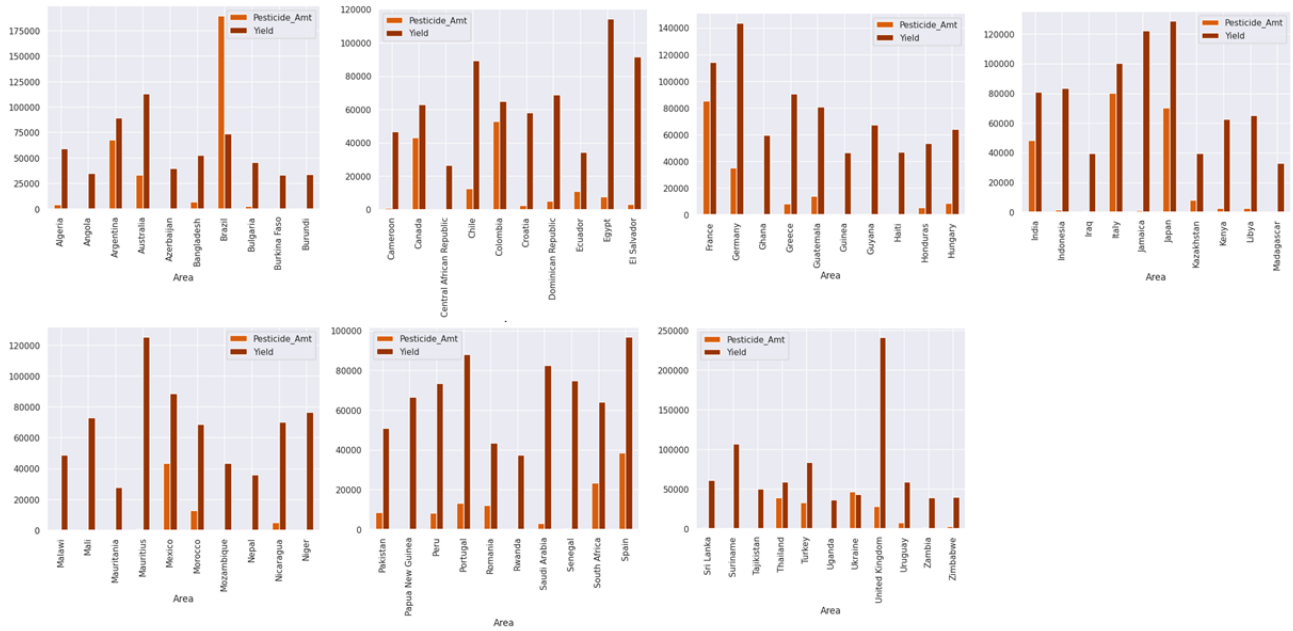
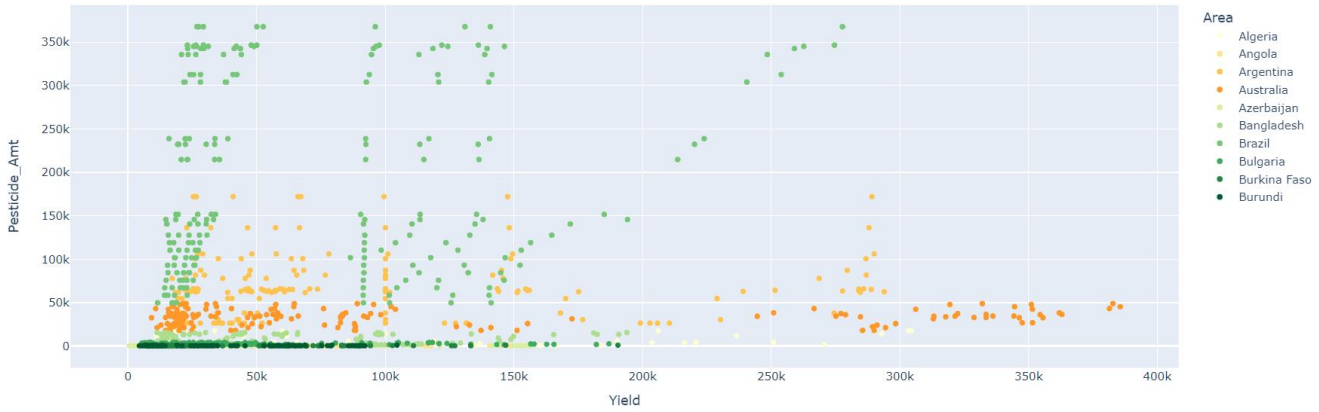
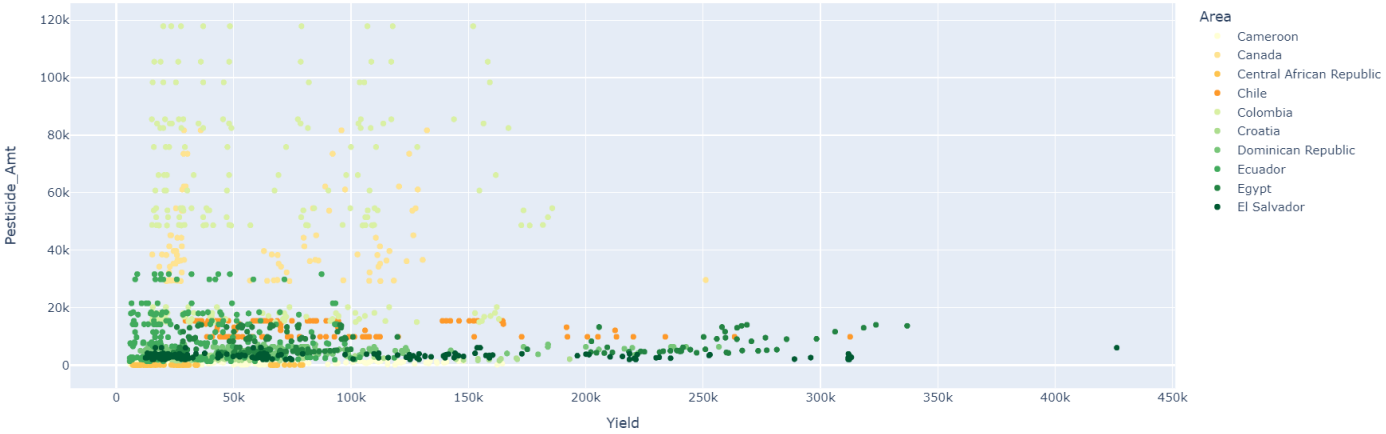


Figure 15: Correlation plot for Yield against Pesticide Usage for each Area


Pesticide_Amt
Yield
Area
Algeria
Angola
Argentina
Australia
Azerbaijan
Bangladesh
Brazil
Bulgaria
Burkina Faso
Burundi


Pesticide_Amt
Yield
Area
Cameroon
Canada
Central African Republic
Chile
Colombia
Croatia
Dominican Republic
Ecuador
Egypt
El Salvador

Pesticide_Amt
Yield
Area
France
Germany
Ghana
Greece
Guatemala
Guinea
Guyana
Haiti
Honduras
Hungary

Pesticide_Amt
Yield
Area
India
Indonesia
Iraq
Italy
Jamaica
Japan
Kazakhstan
Kenya
Libya
Madagascar

Figure 16: Yield obtained against the amount of pesticide used for each unique area

A general pattern could be observed through these correlations. The yield obtained is increased initially with the use of pesticides. Afterwards, it reaches a maximum production value, after which there is a decline in the yield obtained. After this point, the yield does not increase regardless of the amount of pesticides used. Some exceptions were also observed in France and Japan.

For each unique item/crop, the amount of pesticide used was explored. The bar plot is shown below in Figure 17.

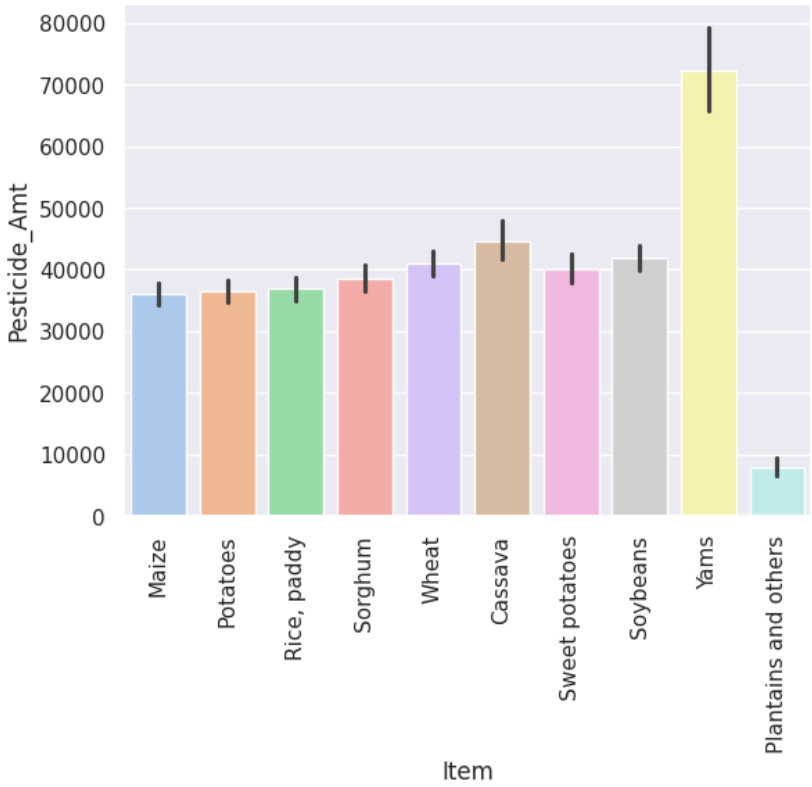


Figure 17: Pesticide usage per Item

It was observed that pesticides were used the highest in the production of yams. While plantains and others used minimal pesticide, the usage remained around 40000 tonnes for the remaining crops.

The yield of each unique item was analysed, giving us the following Figure 18.

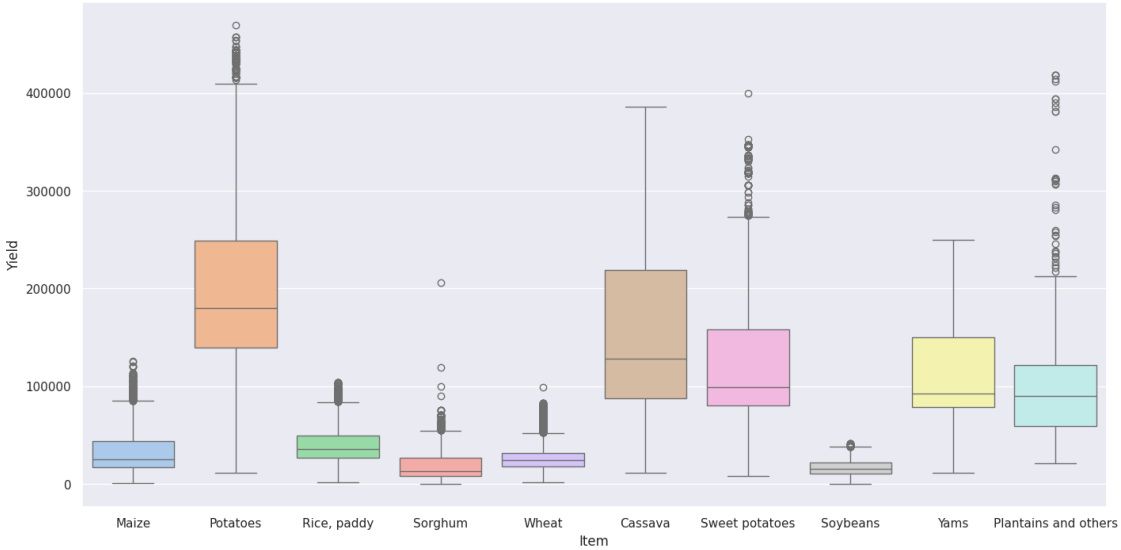


Figure 18: Yield obtained for each unique item

Potatoes and cassava were the crops with the highest yield. Soybeans, wheat and sorghum were amongst the least produced crops.

Different crops thrive in different temperatures. This correlation was analysed, as shown in Figure 19.

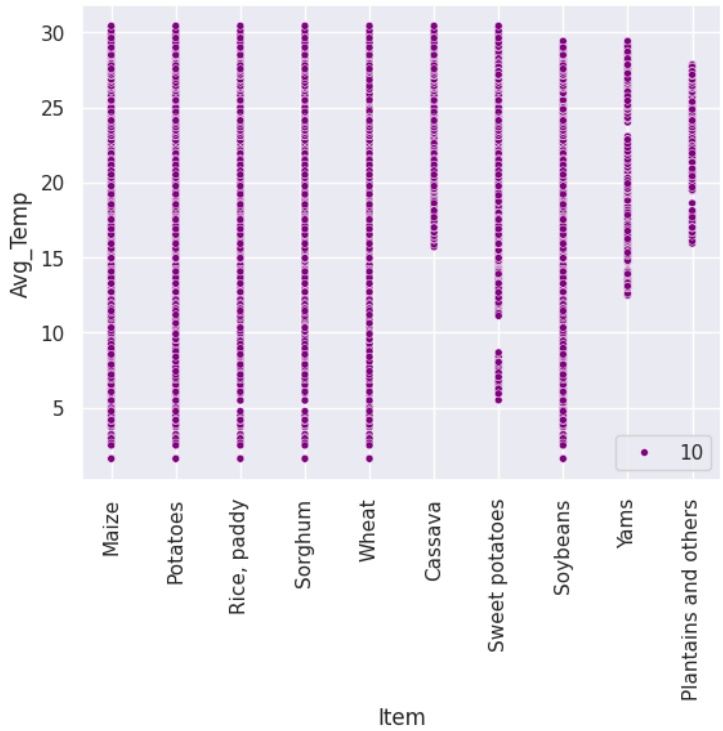


Figure 19: Average temperature against each unique item

It was observed that cassava, yams, plantains and others thrived in comparatively higher temperatures. The remaining crops were able to grow in any temperature range.

The Area/country most suited for each Item/crop was determined. The yield obtained for the respective crop in the area can also be seen in Table 2 below.

	Item	Area	Yield
0	Cassava	India	385818
1	Maize	Tajikistan	125670
2	Plantains and others	Guatemala	418505
3	Potatoes	France	468991
4	Rice, paddy	Australia	103895
5	Sorghum	Algeria	206000
6	Soybeans	Turkey	41609
7	Sweet potatoes	Senegal	400000
8	Wheat	Zambia	99387
9	Yams	Mali	250000

Table 2: Most produced Item and the yield amount for each unique Area/Country

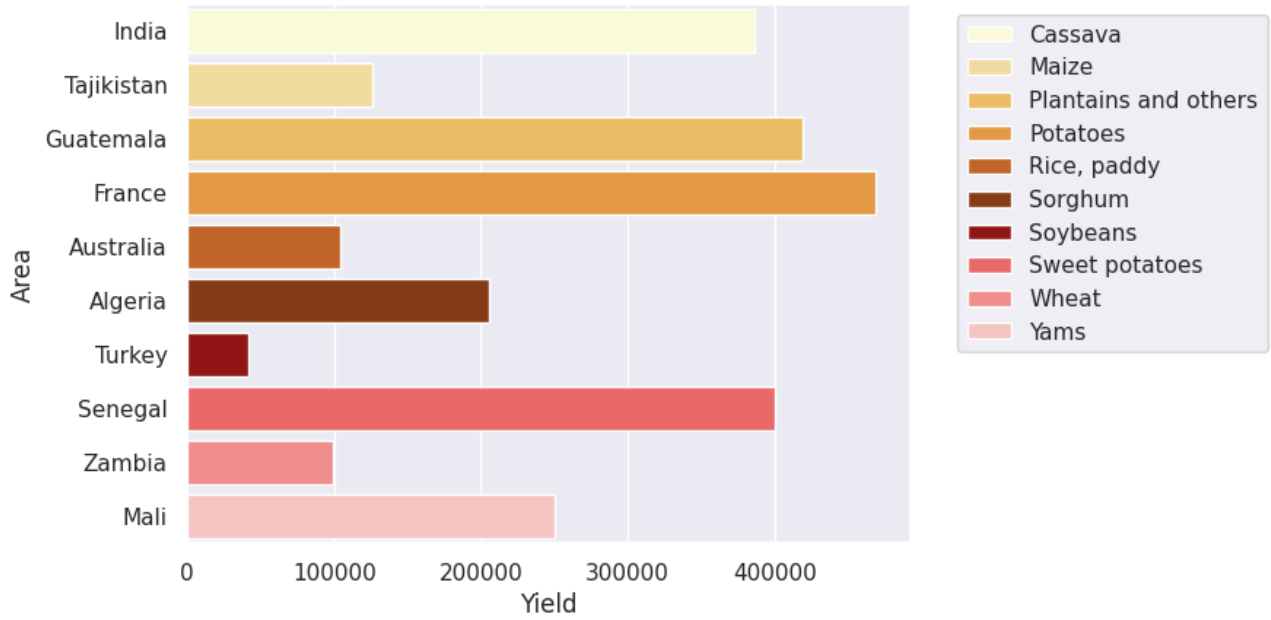


Figure 20: Most produced Item per Area and the yield

The relationship between Yield and the different influencing factors – Pesticide amount used, Average rainfall, average temperature – across the years were analysed and visualised. This was done for each Item/crop separately, colour-coded for easy differentiation. The scatterplot representation is shown below in Figure 21.

Figure 21: Correlation between Yield and various influencing factors

From the study, an initial increase in yield was observed with the usage of pesticides. However, soon the yield attained a maximum level. Any amount of further increase in pesticide usage

showed adverse effects and resulted in a decline of yield obtained. There were some exceptions to this observed pattern in countries such as France and Japan. These were also among the overall top producers. Numerous countries such as Senegal, Suriname, Tajikistan, Mali, Mauritius, Guyana, Algeria, and El Salvador showed impressive yield productions with very minimal use of pesticides. Alternate ways of combating crop issues may have been employed, reducing the need for pesticide usage. On the whole, 30000 tonnes and below usage of pesticides showed to produce high yield and without having adverse effects.

The average yield saw a maximum between 5 and 15 degrees. The yield saw a steady decline with an increase in temperature after 15 degrees. The yield was the lowest when the average temperatures decreased below 5 degrees.

The average rainfall produced the maximum yield between 500 and 1500mm, upwards of which the yield decreased. There was a decline in yield for below 500mm average rainfall as well. The influence of rainfall in crops is not always accurate as irrigation is conducted by the farmers. This is one area that will benefit from on-the-ground sensors for a more accurate prediction.

4 MODEL EVALUATION METHODS

4.1 Mean Squared Error

The Mean Squared Error (MSE) is one of the commonly used evaluation metrics for regression problems. MSE is the average of the squared differences between predicted and expected target values in a dataset. It is calculated as:

MSE = 1 / N * sum for i to N (y_i – yhat_i)^2

where y_i is the i'th expected value in the dataset and yhat_i is the i'th predicted value. In order to give a positive error value, the difference between these two values is squared. A lower value of MSE suggests higher accuracy of the predictive model (Brownlee).

4.2 Root Mean Squared Error

The MSE shows higher error values since it is squared to remove any negative value. The Room Mean Squared Error (RMSE) is calculated to overcome this challenge for display purposes. The formula is:

RMSE = sqrt(1 / N * sum for i to N (y_i – yhat_i)^2)

where y_i is the i'th expected value in the dataset, yhat_i is the i'th predicted value, and sqrt() is the square root function. Similar to MSE, the accuracy of the predictive model is inversely proportional to RMSE (Brownlee).

4.3 Mean Absolute Error

The Mean Absolute Error (MAE) is another common accuracy metric. MSE units match the units of the target value with linear changes. This makes it easier and more intuitive to understand. It is calculated as the average of the absolute error values, with the formula:

MAE = 1 / N * sum for i to N abs(y_i – yhat_i)

where y_i is the i'th expected value in the dataset, yhat_i is the i'th predicted value and abs() is the absolute function. A lower MAE indicates a higher prediction model accuracy (Brownlee).

4.4 R-squared

R-squared is the accuracy metric used for forecast models. It tells us how much of the variability in the outcome variable is explained by the predictor variables in the model. It ranges from 0 to 1. A value of 0 indicates that the model does not explain any variability and 1 indicates that it explains all variability (Bhandari 2024).

4.5 Mean Absolute Percentage Error

The Mean Absolute Percentage Error(MAPE) is another commonly used metric to measure forecast accuracy. It is the sum of the individual absolute errors divided by the demand. In other words, the MAPE is the average of the percentage errors and is calculated by:

$$MAPE = \frac{1}{n} \sum \frac{|e_t|}{d_t}$$

It is not always the best accuracy indicator as it has a tendency to be skewed (Vendeput).

5 FINDINGS AND RESULTS

5.1 Evaluation of the various models implemented

Among the models implemented, **Linear Regression** displayed the least accuracy. It displayed an RMSE score of 79332.95, and an MAE of 60955.32 suggesting considerable errors in the predictions. The model had a R^2 value of 0.0737 indicating only about 7.37% of the variance in the dependent variable is explained by the independent variables

Random Forest was among the better-performing models. It showed an R2 score of 0.985628, meaning approximately 98.56% of the variance in the dependent variable is explained by the independent variables. The RMSE was shown as 9882.02 and MAE as 3480.84. The significantly lower MSE, RMSE, MAE, and MAPE values indicate minimal prediction errors, suggesting robust performance.

The Gradient Boost model showed good accuracy. The R^2 value of 0.831140, indicated that approximately 83.11% of the variance in the dependent variable was explained by the independent variables. However, the higher MSE, RMSE (33872.49), MAE (21184.66), and MAPE values compared to the random forest model suggest slightly inferior performance.

The **XGBoost** and **Decision Tree** models performed similarly with high accuracy. The models had an R^2 value of 0.974317 and 0.976174 respectively. The relatively low MSE, RMSE, MAE and MAPE values indicate precise predictions, suggesting strong performance by both models.

The **KNN** was another model exhibiting low accuracy indicated by an R^2 value of 0.288206. higher MSE, RMSE, MAE, and MAPE values suggest less precise predictions, indicating weaker performance compared to other models.

The **Bagging Regressor** model demonstrates excellent accuracy, with approximately 98.59% of the variance explained by the independent variables. Moreover, the significantly lower MSE, RMSE, MAE, and MAPE values indicate minimal errors in predictions, suggesting robust performance.

	Model	Accuracy	MSE	RMSE	MAE	MAPE	R2_score
0	Linear Regression	0.073724	6293718529.871188	79332.959921	60955.317749	2.419536	0.073724
1	Random Forest	0.985628	97654323.573225	9882.020217	3480.841065	0.102571	0.985628
2	Gradient Boost	0.831140	1147345777.258148	33872.492929	21184.660834	0.596784	0.831140
3	XGBoost	0.974317	174504250.320092	13210.005690	7073.086563	0.198378	0.974317
4	KNN	0.288206	4836388296.236208	69544.146384	47716.358935	1.631186	0.288206
5	Decision Tree	0.976174	161890347.543726	12723.613777	3559.269202	0.096101	0.976174
6	Bagging Regressor	0.985881	95933983.565644	9794.589505	3450.508398	0.101199	0.985881

Figure 22: Evaluation Metrics 1

The **LSTM** model showed an RMSE score of 82539.18 and an MAE value of 64164.13. It indicates that the model's predictions have higher variability or are more sensitive to outliers. The R2 value of 0.068 indicates that only about 68% of the variance in the dependent variable is explained by the model. These metrics suggest that the LSTM model may not be adequately capturing the underlying patterns in the data or making accurate predictions.

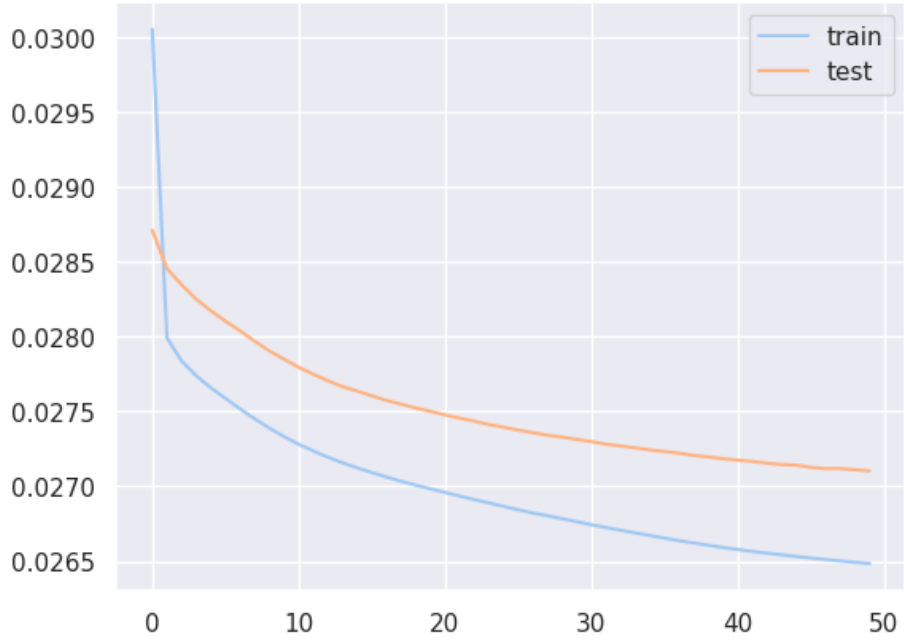


Figure 23: Visualisation for Training and Validation Loss for LSTM

```
Mean Absolute Error (MAE): 64164.12526202987
Mean Absolute Percentage Error (MAPE): 267.03601013237324
Mean Squared Error (MSE): 6812716373.63729
R-squared Value: 0.06805445948511823
Root Mean Squared Error (RMSE): 82539.18083938857
```

Figure 24: Evaluation metrics for the LSTM model

The **CNN** model showed an MAE value of 65225.04, and an RMSE value of 81661.19. This suggests that the model's predictions have higher variability. The R2 value of 0.087 indicates that the model's independent variables do not explain much of the variability observed in the dependent variable. The metrics suggest that the CNN model's performance was suboptimal and a failure to effectively capture the inherent patterns within the dataset or generate precise predictions. It is noted that to enhance the efficacy of LSTM and CNN, further analysis, feature engineering, or model tuning may be necessary.

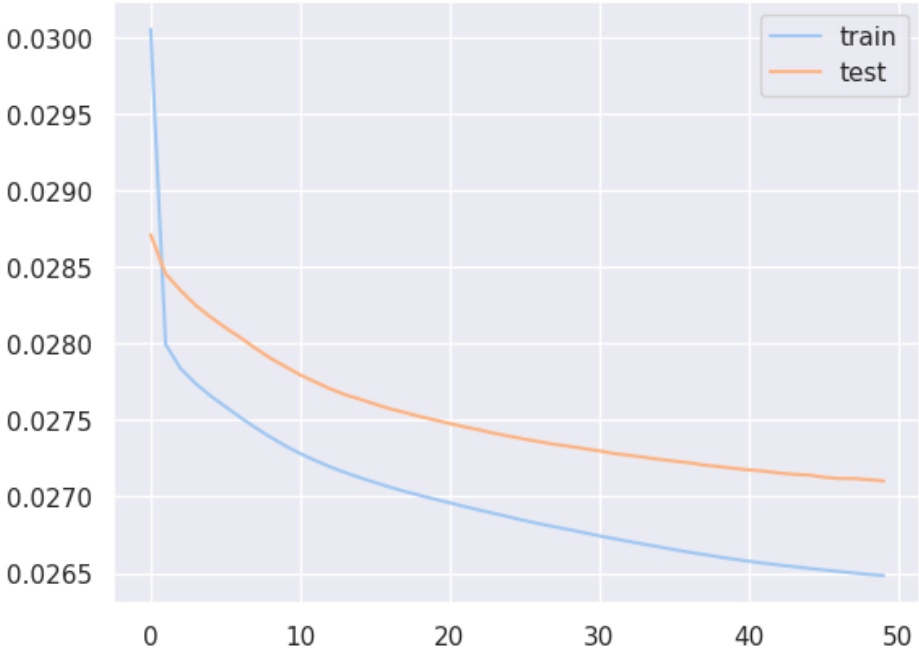


Figure 25: Visualisation for Training and Validation Loss for CNN

```
Mean Absolute Error (MAE): 65225.04095616704
Mean Squared Error (MSE): 6668550062.619164
R-squared Value: 0.08777569008936426
Root Mean Squared Error (RMSE): 81661.19067598246
Mean Absolute Percentage Error (MAPE): 285.75718300913906
```

Figure 26: Evaluation metrics for the CNN model

6 CONCLUSIONS AND FUTURE WORK

6.1 Conclusions

Agriculture and crop management is an essential industry for the survival of humans. The integration of various data analysis methodologies into crop prediction techniques serves as a pivotal means to achieve their primary objective. Moreover, these methods extend their utility to optimizing crop yield, enhancing comprehension of the multifaceted factors influencing yield, and refining the application of pesticides and fertilizers A noteworthy observation from this study is that higher pesticide usage did not always yield maximum product. This underscores the necessity for nuanced analysis and a tailored approach to pesticide application based on crop type and geographical considerations. An average temperature ranging between 5 to 15 degrees Celsius and an average rainfall of 500 to 1500 mm appeared to be most favourable for higher crop yield.

Among the various ML methods implemented, the Bagging Regressor demonstrated the highest accuracy in yield prediction. The model explains approximately 98.59% of the variance in the dependent variable, as accounted for by the independent variables, and showed significantly lower MSE, RMSE, MAE, and MAPE values indicating minimal errors in predictions, suggesting robust performance. Random Forest, XGBoost, and Decision Tree also performed fairly well with 98.56%, 97.43%, and 97.61% respectively. Methods such as LSTM and CNN could enhance the efficacy with further analysis, feature engineering, and/or model tuning.

6.2 Future Work

Several directions for future research presented during this project.

- Exploration of Ensemble Methods: There is potential for further investigation into ensemble methods such as stacking or blending various ML models. This exploration aims to capitalize on the unique strengths of individual algorithms, thus enhancing prediction accuracy.
- Adoption of Advanced Deep Learning Techniques: Advanced deep learning techniques, notably long short-term memory (LSTM) networks and convolutional neural networks (CNNs), hold promise for capturing complex temporal and spatial patterns within crop yield data. Future research endeavours can delve into these methodologies to uncover their efficacy in this domain.
- Integration of Satellite Imagery and Remote Sensing Data: Leveraging high-resolution satellite imagery and remote sensing data offers an opportunity to capture nuanced spatial and temporal variations in crop health and environmental conditions. Further exploration of these datasets can enhance predictive capabilities and refine our understanding of agricultural dynamics.

- Integration of Diverse Data Sources: Investigating methods for integrating diverse datasets from multiple sources, including weather stations, soil sensors, and agricultural machinery, can provide a more comprehensive understanding of crop growth dynamics. Future research can focus on developing robust frameworks for merging and analysing these heterogeneous data streams.
- Development of Dynamic Models: Dynamic modeling approaches that can adapt to changing environmental conditions and incorporate real-time data streams hold immense potential for enhancing prediction accuracy. Future efforts can concentrate on refining these models to provide more accurate and timely predictions for agricultural stakeholders.

By pursuing these avenues of research, the field of crop yield prediction and analysis can advance significantly, leading to more informed decision-making and sustainable agricultural practices.

6.3 Completion of Aims and Objectives

This project aimed to investigate and analyse the diverse factors influencing crop yield. Three primary factors – namely average temperature, average rainfall, and amount of pesticides used - were examined using the collected data. The objective of implementing and comparing regression models for predicting crop yield was successfully achieved. However, there is scope for further research to explore additional factors and conduct a more nuanced study of each factor. A detailed analysis of the influencing factors and their integration into ML techniques could enhance the efficiency and accuracy of the models.

Word count (main body of the report): 5985

7 REFERENCES

Ansarifar, J. W. L. &. A. S., 2021. An interaction regression model for crop yield prediction. *Scie Rep,* p. 17754.

Bhandari, A., 2024. *Key Difference between R-squared and Adjusted R-squared for Regression Analysis.* [Online] Available at: https://www.analyticsvidhya.com/blog/2020/07/difference-between-r-squared-and-adjusted-r-squared/

Bhattacharyya, B. &. B. R. &. C. D., 2021. LINEAR REGRESSION MODEL TO STUDY THE EFFECTS OF WEATHER VARIABLES ON CROP YIELD IN MANIPUR STATE. *International Journal of Agricultural and Statistics Sciences,* pp. 317-320.

Brownlee, J., n.d. *Machine Learning Mastery.* [Online] Available at: https://machinelearningmastery.com/regression-metrics-for-machine-learning/

Dharani, M. T. R. N. P. K. P. a. S. S., 2021. Review on crop prediction using deep learning techniques. *Journal of Physics: Conference Series,* 1767(1), p. 012026.

Dhivya Elavarasan, D. R. V. V. S. A. Y. Z. K. S., 2018. Forecasting yield by integrating agrarian factors and machine learning models: A survey. *Computers and Electronics in Agriculture,* pp. 257-282.

Dokoohaki, H. K. M. M.-F. R. M. F. a. H. G., 2021. A comprehensive uncertainty quantification of large-scale process-based crop modeling frameworks. *Environmental Research Letters,* 16(8), p. 084010.

Fahad, S. H. S. B. A. e. a., 2015. Potential role of phytohormones and plant growth-promoting rhizobacteria in abiotic stresses: consequences for changing environment. *Environmental Science and Pollution Research,* Volume 22, pp. 4907-4921.

FAO, IFAD, UNICEF, WFP and WHO, 2023. *The State of Food Security and Nutrition in the World 2023.,* Rome, Italy: FAO ; IFAD ; UNICEF ; WFP ; WHO.

Feng, X. P. A. K. G. X. L. X. L. a. X. Z., 2017. Growth, Root Compensation and Ion Distribution in Lycium Chinense under Heterogeneous Salinity Stress. *Scientia Horticulturae,* pp. 24-32.

Garcia, F. A. S. O. J. H. A. a. A. P., 2012. Pesticides: classification, uses and toxicity. Measures of exposure and genotoxic risks. *International Journal of Environmental Science and Toxicology,* pp. 279-293.

Gavahi, K. A. P. &. M. H., 2021. DeepYield: A combined convolutional neural network with long short-term memory for crop yield forecasting. *Expert Systems With Applications,* pp. 184-115511.

Hossein Zare, T. K. W. J. I. W. N. S. G. T. S., 2024. Within-season crop yield prediction by a multi-model ensemble with integrated data assimilation. *Field Crops Research,* p. 109293.

IBM, n.d. [Online] Available at: https://www.ibm.com/topics/random-forest#:~:text=Random%20forest%20is%20a%20commonly,both%20classification%20and%20regression%20problems.

J. Huang, J. G.-D. H. H. H. M. Q. W. P. L. S. L. Z. C. J.-H. X. Y. W. F. Z. J. W. X. X., 2019. Assimilation of remote sensing into crop growth models: Current status and perspectives. *Agricultural and Forest Meteorolgy,* Volume 276-277, p. 107609.

J. You, X. L. M. L. D. L. S. E., 2017. *Deep Gaussian process for crop yield prediction based on remote sensing data.* California USA, s.n.

Jin, X. Y. G. X. X. Y. H. F. H. L. Z. S. J. Z. C. a. L. Y., 2015. Combined multi-temporal optical and radar parameters for estimating LAI and biomass in winter wheat using HJ and RADARSAR-2 data. *Remote Sensing,* 7(10), pp. 13251-13272.

Kim, N. a. L. Y., 2016. Machine Learning Approaches to Corn Yield Estimation Using Satellite Images and Climate Data: A Case of Iowa State: A Case of Iowa State. *한국측량학회지,* pp. 383-390.

Koirala, A. W. K. W. Z. a. M. C., 2019. Deep learning–Method overview and review of use for fruit detection and yield estimation. *Computers and electronics in agriculture,* Volume 162, pp. 219-234.

Kumar, S. P. N. a. H. P. M., 2022. *Soil Quality Identifying and Monitoring Approach for Sugarcane Using Machine Learning Techniques.* Mandya, India, IEEE, pp. 1-5.

L. Kouadio, R. D. V. B. J. A. S. M. V. N., 2018. Artificial intelligence approach for the prediction of robusta coffee yield using soil fertility properties. *Comput. Electron. Agric.,* pp. 324-338.

Lal, R., 2009. Soil degradation as a reason for inadequate human nutrition. *The Science, Sociology and Economics of Food Production and Access to Food,* pp. 45-57.

Lee, S. J. Y. S. S. a. L. B., 2019. A self-predictable crop yield platform (SCYP) based on crop diseases using deep learning. *Sustainability,* 11(13), p. 3637.

M. Maimaitijiang, V. S. P. S. S. H. F. E. F. F., 2020. Soybean yield prediction from uav using multimodal data fusion and deep learning. *Remote Sens. Environ.,* p. 237.

MathWorks, n.d. [Online] Available at: https://uk.mathworks.com/help/stats/what-is-linear-regression.html

N. Chergui, T. K. M. M., 2020. *The impact of data analytics in digital agriculture: A review.* Tunis, Tunisia, s.n., pp. 1-13.

N. Kim, K. H. N. P. J. C. S. H. Y. L., 2019. A comparison between major artificial intelligence models for crop yield prediction: Case study of the midwestern United States, 2006–2015. *ISPRS International Journal of Geo-Information,* p. 5.

Popp, J. P. K. &. N. J., 2013. Pesticide productivity and food security. A review. *Agronomy for Sustainable Development,* pp. 243-255.

Prasad, N. P. N. &. D. A., 2021. Crop yield prediction in cotton for regional level using random forest approach. *Spatial Information Research,* pp. 195-206.

Ramos, P. P. F. M. E. &. O. C., 2017. Automatic fruit count on coffee branches using computer vision. *Computers and Electronics in Agriculture,* pp. 9-22.

S. Gyamerah, P. N. D. I., 2020. Probabilistic forecasting of crop yields via quantile random forest and epanechnikov kernel function. *Agric. For. Meteorol.,* p. 280.

Shaikh, T. M. W. R. T. e. a., 2022. Machine Learning for Smart Agriculture and Precision Farming: Towards Making the Fields Talk. *Arch Computat Methods Eng,* p. 4557–4597.

Sun, J. D. L. S. Z. S. Y. &. L. Z., 2019. County-Level Soybean Yield Prediction Using Deep CNN-LSTM Model. *Sensors,* p. 4363.

Tudi, M. D. R. H. W. L. L. J. S. R. C. D. C. C. &. P. D. T., 2021. Agriculture Development, Pesticide Application and Its Impact on the Environment. *International Journal of Environmental Research and Public Health,* p. 1112.

Van Klompenburg, T. K. A. a. C. C., 2020. Crop yield prediction using machine learning: A systematic literature review. *Computers and Electronics in Agriculture,* Volume 177, p. 105709.

Vendeput, N., n.d. Data Science for Supply Chain Forecasting. In: s.l.:s.n.

X. Jin, L. K. Z. L. H. F. X. X. G. Y. J. W., 2018. A review of data assimilation of remote sensing and crop models. *European Journal of Agronomy,* Volume 92, pp. 141-152.

Xiong, L. &. Z. K., 2002. Molecular and genetic aspects of plant responses to osmotic stress. *Plant, Cell & Environment,* pp. 131-139.

Zhang, Q. L. Y. G. C. C. Y. a. Y. H., 2020. Applications of deep learning for dense scenes analysis in agriculture: A review. *Sensors,* 20(5), p. 1520.

8 APPENDICES

8.1 APPENDIX A

Project Proposal Form

Degree Title: **MSc Data Science and Artificial Intelligence**	**Student's Name: Sushma Nair Achnath**
	Supervisor's Name: Dr Emili Balaguer-Ballester
	Project Title/Area: Crop Yield Prediction using ML Techniques

Section 1: Project Overview

1.1 Problem definition:

The many factors that affect crop yield are collected and processed using Machine Learning algorithms to ensure continued and maximized crop yield is essential to meet the growing nutritional needs of our growing population.

1.2 Project description:

The project will process data on various factors such as soil type & composition, climate, humidity and temperature, and seed quality. These will be categorized in multiple ways and will try to form a model and guide to predict what factors will be favourable for maximum yield for any given crop.

1.3 Background:

With the growing population globally, food security is a big concern that necessitates accurate and extensive data-driven solutions to address its complex challenges. There is a need to maintain ample supply of healthy food for global sustenance. We need to ensure this for global wellbeing simultaneously preserving the Earth's ecosystems. Addressing this presents a huge challenge, considering that food security is of a complex and multifaceted nature. (Shoaib, 2023) Multiple AI tools have been developed in agriculture to help farmers achieve exact and controlled cultivation. (Jorvekar, 2023) This project will take into account various external factors such as temperature, precipitation, soil moisture, terrain, climate, soil composition. It will pre-process and categorize them in different ways. (Van Klompenburg, 2020)Then we will deploy ML techniques to predict the crop yield in relation to variations in these factors. (Houshmandfar) We will also introduce an optimization algorithm to maximize yield for each crop by bringing about necessary changes in these external factors. (2)

1.4 Research Questions

Question 1: How effective are Machine Learning/Deep Learning techniques in processing and predicting

crop yield efficiently?

Question 2: Which ML model will best predict crop yield considering the various external factors affecting yield?

Question 3: Can we build an optimization model for ensuring maximum crop yield?

1.5 Aims and objectives:

The aim of this project is to use ML techniques to efficiently predict crop yield and develop a model to optimize crop yield.

Section 2: Artefact

2.1 What is the artefact that you intend to produce?

A novel crop yield prediction model using Python and determine how to optimize the various contributing external factors that will enable to maximize yield for each crop.

2.2 How is your artefact actionable?

We are able to measure/monitor the various external factors that influence a crop yield. This project will enable users to predict the difference in yield in relation to changes in these factors. This will be done by processing extensive datasets collected across many years. It will help user to consider and change these factors optimally to ensure maximum yield possible, thus helping to meet the food supply needs globally.

Section 3: Evaluation

3.1 How are you going to evaluate your project artefact?

This work will address limitations that exist in current crop yield prediction methods. It will consider multiple factors and streamline which contribute the most towards a yield and discuss methods to maximize the same. The aim is to develop an algorithm so that stakeholders find it easy to use and optimize their crop.

3.2 How does this project relate to your MSc Programme and your degree title outcomes?

This project is related to MSc Data Science and Artificial Intelligence by enabling me to implement the knowledge I have acquired to process and predict crop yield efficiently. This will implement Deep learning and Machine Learning techniques to process the factors affecting crop yield and predict their effects on the same. It will make use of optimization techniques to guide farmers to maximize their yield.

3.3 What are the risks in this project and how are you going to manage them?

There is extensive data from across the globe related to crop yield. Though using as varied data as possible to determine the various relations and influence of factors will help in accurately predicting crop yield, this has to be done in a timely manner. Choosing what factors to consider to best gather this data is essential. Multiple AI algorithms need to be tested out to determine the best solution.

Section 4: References

4.1 Please provide references if you have used any.

Shoaib, M.R., Emara, H.M. and Zhao, J., 2023. Revolutionizing Global Food Security: Empowering Resilience through Integrated AI Foundation Models and Data-Driven Solutions. *arXiv preprint arXiv:2310.20301*.

Jorvekar, P.P., Wagh, S.K. and Prasad, J.R., 2023. Crop yield predictive modeling using optimized deep convolutional neural network: An automated crop management system. *Multimedia Tools and Applications*, pp.1-28.

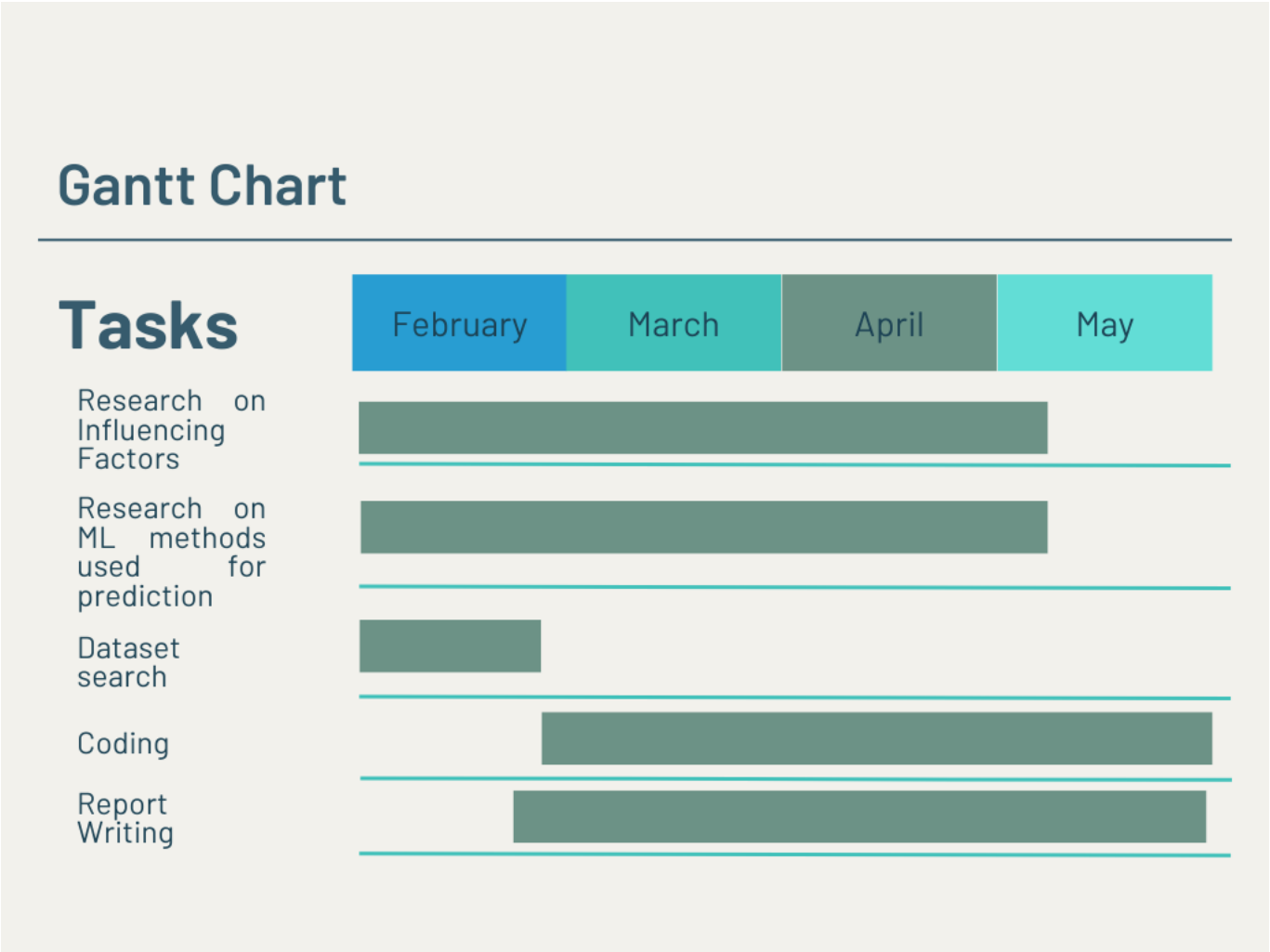
Houshmandfar, A., Can Machine Learning with Climate Data Alone Achieve High Yield Prediction Accuracy in Australian Dryland Cropping Systems?. *Available at SSRN 4610893*.

Van Klompenburg, T., Kassahun, A. and Catal, C., 2020. Crop yield prediction using machine learning: A systematic literature review. *Computers and Electronics in Agriculture*, *177*, p.105709.

Section 5: Academic Practice and Ethics

5.1 Have you made yourself familiar with, and understand, the University guidance on referencing and plagiarism?	**Yes**
5.2 Do you acknowledge that this project proposal is your own work and that it does not contravene any academic offence as specified in the University's regulations?	**Yes**
5.3 Have you submitted online ethics checklist to your supervisor?	**Yes**
5.4 Has the checklist been approved by your supervisor?	**Yes**

Section 6: Proposed Plan (please attach your <u>Gantt chart</u> below)



8.2 APPENDIX B

Bournemouth University

Research Ethics Checklist

About Your Checklist	
Ethics ID	53039
Date Created	16/11/2023 10:22:35
Status	Approved
Date Approved	17/11/2023 20:04:21
Risk	Low

Researcher Details	
Name	Sushma Achnath
Faculty	Faculty of Science & Technology
Status	Postgraduate Taught (Masters, MA, MSc, MBA, LLM)
Course	MSc Data Science & Artificial Intelligence

Project Details	
Title	Crop Yield Prediction using ML Techniques
Start Date of Project	28/01/2024
End Date of Project	29/03/2024
Proposed Start Date of Data Collection	29/01/2024
Supervisor	Emili Balaguer-Ballester
Approver	Emili Balaguer-Ballester
Summary - no more than 600 words (including detail on background methodology, sample, outcomes, etc.)	
The project will process data on various factors such as soil type & composition, climate, humidity and temperature, seed quality. These will be categorized in multiple ways and will try to form a model and guide to predict what factors will be favourable for maximum yield for any given crop.	

Filter Question: Does your study involve the use or re-use of data which will be obtained from a source other than directly from a Research Participant?

Additional Details	
Please describe the data, its source and how you are permitted to use it	Data will be crop yield details accessed from websites such as Kaggle.

Research Data

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Will identifiable personal information be collected, i.e. at an individualised level in a form that identifies or could enable identification of the participant?	No
Will research outputs include any identifiable personal information i.e. data at an individualised level in a form which identifies or could enable identification of the individual?	No

Storage, Access and Disposal of Research Data	
Where will your research data be stored and who will have access during and after the study has finished.	
The data is readily available online. It will remain to be so.	
Once your project completes, will your dataset be added to an appropriate research data repository such as BORDaR, BU's Data Repository?	Yes

Final Review	
Are there any other ethical considerations relating to your project which have not been covered above?	No

Risk Assessment	
Have you undertaken an appropriate Risk Assessment?	Yes

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8.3 APPENDIX C

Masters Project First Progress Review

Activity: First Progress Review (FEEDBACK ONLY)

Course: Individual Masters Project (Feb) 2024

Name: Sushma Achnath

Criteria	Yes	To some extent	No
Definition of the problem - Has the problem been defined, has the artefact been identified and have objectives been set?	Yes ✓	To some extent	No
Criterion Feedback Is a well-defined regression problem.			
Review of literature and related work - Is there evidence of appropriate research?	Yes ✓	To some extent	No
Criterion Feedback Specialized sources identified, suggest structuring the chapter in liking themes to create a narrative, as we discussed.			
Methodology and Artefact - Is there evidence of appropriate analysis of the problem and design of a solution and appropriate evaluation?	Yes ✓	To some extent	No
Dissertation - Have sections of the dissertation been written and has the Supervisor seen these?	Yes ✓	To some extent	No

Criteria	Yes	To some extent	No
Criterion Feedback Please see previous comments. Suggest also to work on the template for Chapter 1 and consider our discussion on objectives and novelty.			
Planning & Progress - Is there an acceptable plan for this project and is it being followed?	Yes ✓	To some extent	No
Criterion Feedback Maybe mention it in Ch 3.			
Proposal & Online Ethics Checklist - Are proposal and ethic checklist submitted? Are they approved?	Yes ✓	To some extent	No
Form completed with student	Yes ✓		No

Overall Score

Satisfactory	Requires minor improvement	Requires major improvement (at some risk of failing the project)	Unsatisfactory (at high risk of failing the project)	Invalid - Proposal not approved	Invalid - Proposal not submitted	Invalid - Ethics checklist not approved	Invalid - Ethics checklist not submitted

8.4 APPENDIX D

LIST OF CONTENTS OF THE LARGE FILE SUBMITTED

1. Dataset used – “FinalCropYieldData”

2. Codes

   2.1 Code for Data Integration – “crop_yield_prediction_data_integration”

   2.2 Implementation of analysis and prediction – “crop_yield_prediction_Final”

8.5 APPENDIX E

The project was done in a system with the following specifications:

- Device name DESKTOP-0GTVVP6
- Processor Intel(R) Core(TM) i5-8250U CPU @ 1.60GHz 1.80 GHz
- Installed RAM 8.00 GB (7.90 GB usable)
- System type 64-bit operating system, x64-based processor

The code was written in Python using Google Colab.